

F3: BOUNDED-QUERY HARDNESS WITH UNRESTRICTED QUERY SIZE (COMPILE TEST)

Anonymous

ABSTRACT

Compile test for `results/F3_hardness_full.tex`. Not a paper section.

Design constants. For an integer $K \geq 3$ and a real $\eta > 1$ put

$$\begin{aligned} k_1 &= \eta(K-1) + 1, & q &= 1 - \frac{1}{k_1}, & \nu &= \frac{\eta}{\eta-1}, \\ j &= [K+1 - \lceil \eta \rceil]_0^K, & m^* &= \lceil \eta K \rceil - 1, & T &= j + m^*, \\ D &= \max \left\{ \frac{q^j}{K\eta}, \frac{q^j (\nu^{m^*}/K-1)}{\eta(\nu^{m^*}-1)-m^*} \right\}, & W_K(\eta) &= 1 - q^j + (K-j)D. \end{aligned}$$

Lemma 1 (Hypergeometric upper tail, explicit, any query size). *Let O be a uniformly random K -subset of N , $|N| = n$, let $S \subseteq N$ be fixed with $|S| = s \leq n$, and let $1 \leq m \leq K$. Then*

$$\Pr[|S \cap O| \geq m] \leq \binom{K}{m} \prod_{i=0}^{m-1} \frac{s-i}{n-i} \leq \binom{K}{m} \left(\frac{s}{n}\right)^m \leq \frac{1}{m!} \left(\frac{Ks}{n}\right)^m.$$

In particular $\Pr[|S \cap O| \geq 2] \leq K^2 s^2 / (2n^2)$.

The family. With the constants above define, for $0 \leq x \leq n-K$ and $0 \leq y \leq K$,

$$r_x = \begin{cases} q^x, & x \leq j \\ q^j - (x-j)D, & j < x \leq T \\ 0, & x > T \end{cases} \quad g_x = \begin{cases} q^x/K, & x \leq j \\ \nu^{x-j} q^j / K - \eta D(\nu^{x-j} - 1), & j < x \leq T \\ 0, & x > T, \end{cases}$$

$$F_O(S) = 1 - r_x + \mathbf{1}\{y \geq 1\} g_x + \max(y-1, 0) \frac{r_x - g_x}{K-1}, \quad \hat{G}_s = \frac{1}{\sqrt{\eta}} \sum_{x < s} g_x,$$

$$G_O(S) = \begin{cases} \hat{G}_{x+y}, & y \leq 1 \\ \hat{G}_{x+1} + \sqrt{\eta} (F_O(S) - F_O(S_{x,1})), & y \geq 2, \end{cases}$$

where $S_{x,1}$ is any set with the same x and $y=1$. Set $f = F_O$ and $\tilde{f} = G_O$.

Lemma 2 (The family is admissible). *F_O is monotone and submodular, $F_O(\emptyset) = 0$, $F_O(O) = 1$ and $F_O(S) \leq 1$ for every $|S| \leq K$, so $\text{OPT} = F_O(O) = 1$. The pair (F_O, G_O) has single-element error exactly η (that is, $\delta = 0$: no error inflation is needed). Moreover $F_O(S) = 1 - r_{|S|}$ whenever $S \cap O = \emptyset$, and $F_O(T) \leq 1 - r_K = W_K(\eta)$ for every T with $|T| \leq K$ and $T \cap O = \emptyset$.*

Lemma 3 (The predictor is O -oblivious off a bounded window). *$G_O(S) = \hat{G}_{|S|}$ holds whenever $|S \cap O| \leq 1$, and also whenever $|S \setminus O| > T$. Consequently $G_O(S) \neq \hat{G}_{|S|}$ forces both $|S \cap O| \geq 2$ and $|S| \leq T + K$.*

Lemma 4 (Valuation on the good event). *Let $\mathcal{A}$ be deterministic, make at most Q queries to $\tilde{f}$ of arbitrary size, and output T with $|T| \leq K$. Run $\mathcal{A}$ against the canonical oracle $S \mapsto \hat{G}_{|S|}$ and let $S_1, \dots, S_Q, T_0$ be the resulting (fixed, O -free) transcript and output. If*

$$\text{for every } i: \quad |S_i \cap O| \leq 1 \text{ or } |S_i \setminus O| > T, \quad \text{and} \quad T_0 \cap O = \emptyset,$$

then $\mathcal{A}$ run on (F_O, G_O) outputs T_0 and $F_O(T_0)/F_O(O) \leq W_K(\eta)$.

Theorem 5 (Bounded-query hardness, unrestricted query size). *Let $K \geq 3$, $\eta > 1$, and let $n \geq 4K(T + K)$ (which implies $n \geq 8K^2$). Let $\mathcal{A}$ be any deterministic algorithm that makes at most*

$$Q \leq \frac{n^2}{2K^2(T + K)^2} \quad \left(\text{and } T + K \leq (2 + \eta)K + 1, \text{ so } Q = \Omega(n^2 / ((2 + \eta)^2 K^4)) \right)$$

queries to $\tilde{f}$, of arbitrary size, and outputs a set of size at most K . Then there is a K -set $O \subseteq N$ for which the instance $(f, \tilde{f}) = (F_O, G_O)$ is monotone submodular with single-element error exactly η , has $O^ = O$, and*

$$\frac{f(\mathcal{A})}{f(O^*)} \leq W_K(\eta) = V_j(\eta) + (K - j) \left(D - \frac{q^j}{K\eta} \right).$$

Remark 6 (What the constant is). $W_K(\eta) = V_j(\eta) + (K - j) \left(D - q^j / (K\eta) \right)$, where V_j is the closed form of the reduced (predictive-greedy) LP of R10, and the excess term is positive but decays very fast in K : at $\eta = 2$ it is $3.2 \cdot 10^{-4}, 5.9 \cdot 10^{-7}, 4.5 \cdot 10^{-12}, 5.2 \cdot 10^{-22}$ for $K = 4, 8, 16, 32$. Also $L_K(\eta) \leq V_j(\eta) \leq W_K(\eta) < U_K(\eta)$ and $W_K(\eta) \rightarrow 1 - e^{-1/\eta}$ as $K \rightarrow \infty$.

Corollary 7 (Randomized version). *With the parameters of Theorem 5, let $\mathcal{R}$ be any randomized algorithm with the same query budget and no query-size restriction. Then*

$$\min_O \frac{\mathbb{E}[f_O(\mathcal{R})]}{f_O(O^*)} \leq W_K(\eta) + \varepsilon_n, \quad \varepsilon_n = \frac{K}{n} + \frac{Q K^2 (T + K)^2}{2n^2},$$

and $\varepsilon_n \rightarrow 0$ as $n \rightarrow \infty$ with K, η fixed and $Q = o(n^2 / K^2 (T + K)^2)$.

REFERENCES